



B.2 Greenhouse effect

The particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

All objects emit thermal radiation with intensity per unit area given by the Stefan-Boltzmann law, $P/A = \sigma T^4$, and the peak wavelength of emission follows Wien's law, $\lambda_{\text{max}} \cdot T = \text{constant}$, so hotter objects emit shorter-wavelength radiation. The Sun (very hot) emits mainly visible/short-wave radiation that passes through Earth's atmosphere, while Earth (cooler) re-emits mainly infrared (long-wave) radiation. Greenhouse gases (CO_2 , H_2O , CH_4 , etc.) are largely transparent to incoming shortwave solar radiation but absorb and re-emit outgoing longwave infrared radiation, trapping thermal energy and warming the surface; the balance of incoming and outgoing radiation determines Earth's equilibrium temperature, and albedo describes the fraction of incident radiation reflected rather than absorbed.

Worked example: follow the reasoning

Estimate the peak wavelength of radiation emitted by the Sun's surface ($T \approx 5800 \text{ K}$) using Wien's law, $\lambda_{\text{max}} \cdot T = 2.90 \times 10^{-3} \text{ m} \cdot \text{K}$.

$$\lambda_{\text{max}} = (2.90 \times 10^{-3}) / 5800 = 5.00 \times 10^{-7} \text{ m}.$$

Answer: $\lambda_{\text{max}} \approx 5.00 \times 10^{-7} \text{ m}$ (500 nm, visible light).

Quick check · multiple choice

Why does Earth's surface mainly emit infrared radiation while the Sun mainly emits visible light?

A Earth is farther from the Sun

B Earth has a much lower surface temperature, so by Wien's law its peak emission wavelength is longer

C The atmosphere converts visible light to infrared

D Earth's surface absorbs all visible light and reflects only infrared

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A planet has surface temperature 288 K and radius $6.4 \times 10^6 \text{ m}$, and may be modelled as a black body radiating uniformly over its surface area ($A = 4\pi r^2$). [7]
- (a) Calculate the surface area of the planet.
- (b) Using the Stefan-Boltzmann law ($\sigma = 5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$), calculate the total power radiated by the planet.
- (c) Explain, using the concept of the greenhouse effect, why the planet's actual measured surface temperature is higher than the value predicted by a simple radiative balance model with no atmosphere.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $A = 4\pi r^2 = 4\pi(6.4 \times 10^6)^2 = 5.15 \times 10^{14} \text{ m}^2$.

(b) $P = \sigma T^4 A = 5.67 \times 10^{-8} \times 288^4 \times 5.15 \times 10^{14}$.

$288^4 = 6.88 \times 10^9$. $P = 5.67 \times 10^{-8} \times 6.88 \times 10^9 \times 5.15 \times 10^{14} \approx 2.01 \times 10^{17} \text{ W}$.

(c) Greenhouse gases in the atmosphere absorb outgoing longwave infrared radiation emitted by the surface and re-emit part of it back downward, reducing net energy loss to space; this raises the equilibrium surface temperature above the value expected for a bare planet with no atmosphere.

Quick check answer

B. By Wien's law $\lambda_{\text{max}} \propto 1/T$; Earth's much lower temperature shifts its peak emission to longer (infrared) wavelengths compared to the hot Sun's visible-peaked emission.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.